

Brandon Sosa

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EDUCATION

2021–Present Ph.D. in Earth System Science, Florida International University, Def.: Fall 2026

2017–2020 B.S. in Aquatic Sciences: Marine and Freshwater, Purdue University

RESEARCH AND EXPERIENCE

2024–Present Research Analyst

The Everglades Foundation

- Developed a valuation model estimating \$1 trillion in Everglades ecosystem services over 50 years, in collaboration with Earth Economics, to guide policy and restoration investments
- Applied economic modeling and GIS to map service values across Everglades habitats, covering real estate, recreation, storm protection, and biodiversity
- Building an ecosystem benefit model workflow to spatially transfer wetland service values, using LLM-assisted literature extraction to compile primary valuation estimates

2021–Present Doctoral Candidate

Florida International University, Department of Earth and Environment

- Designed and ran quarry-based mariculture trials of the Caribbean King Crab to evaluate herbivore supply for reef restoration
- Built an agent-based model of Florida reef dynamics to test grazer stocking scenarios and identify cost-effective management interventions
- Designed and deployed a statewide discrete choice experiment to estimate willingness-to-pay for reef restoration outcomes under climate uncertainty

Summer 2025 NSF International Research Experiences for Students (IRES)

Transdisciplinary Coral Reef Science, Australia

- Trained in coral reef field methods at Lizard Island Research Station and research-to-policy translation through social science workshops at James Cook University and research visit to the Australian Institute of Marine Science (AIMS)

2022–2024 Research Assistant

Tropical Marine Ecology Lab, Florida International University

- Led and assisted in field operations of the restoration of nearly 2,000 corals and coordinated crab stocking treatments across multiple patch reefs in the Lower FL Keys
- Monitored restoration success through fish video analysis, benthic cover estimation, and coral outplant fate tracking
- Supported lab projects on sponge filtration, seagrass surveys, and ghost-fishing mitigation through lobster trap accessory trials

2021–Present Collaborative Research Projects

Biscayne National Park Marine Reserve Economic Impact Study (2025–Present)

- Contributing to economic impact assessment of a proposed no-take reserve zone

- Co-developing a scenario planning tool to project outcomes across reserve configurations for park managers, in collaboration with the National Parks Conservation Association and FIU

Coral In Situ Metabolism (CISME) Field Campaign, Lower Florida Keys (2025-Present)

- Served as science diver measuring in situ respiration and photosynthesis of calcifying reef organisms via benthic flux chambers, in collaboration with Nova Southeastern University, Mote Marine Lab, and Georgia Southern University

Shedd Aquarium Research Cruise, R.V. Coral Reef II (July 2023)

- Conducted reef assessments at 13 sites documenting invertebrate grazer abundance, in collaboration with Shedd Aquarium, NOAA, FIU, and UT Austin

Everglades Restoration Cost-Benefit Analysis, Florida Coastal Everglades LTER (2022)

- Conducted benefit-cost analysis quantifying a 3:1 return on investment for restoration projects (CEPP), influencing regional policy through stakeholder presentations

2019–2021 Fisheries Technician

Höök Lab, Purdue University

- Evaluated trophic effects of microplastics from plankton to fish using microscopy, PCR for fish sex determination, and ImageJ for morphometric analysis
- Acknowledged contributor in [Malinowski et al., 2023](#) resulting from this research

Summer 2020 Marine Biology Practicum, Purdue University

- Ten-day field course at Seahorse Key Marine Laboratory, University of Florida; conducted seine netting, species identification, and oyster reef surveys on Florida's Gulf

PUBLICATIONS

Sosa, B. M., Malinowski, C.R. and Höök, T.O. (2026). “Implications of Maternal Toxicant Effects on Size-Dependent Fisheries’ Management: A Meta-Analysis.” *Fish and Fisheries* 1–17. [Q1-ranked journal by Scopus] <https://doi.org/10.1111/faf.70089>.

Manuscripts in Preparation or in Review

Sosa, B., Jarrett, M., and Butler, M.J. Quarry-based mariculture of the herbivorous Caribbean King Crab in support of coral reef restoration. *In review* at Aquaculture Reports. [SSRN](#)

Sosa, B., and Bhat, M. Restoring Reefs: Willingness-to-Pay for Coral Reef Restoration Outcomes Under Uncertain Climate Futures. *In prep.*

Bhat, M., Chabba, M., **Sosa, B.**, Vorseth, C., and Fache, A. Everglades Restoration Under Climate Change: A Benefit-Cost Analysis Across Future Scenarios. *In prep.*

Dilworth, J., Blanco Pimentel, M., Cunning, R., Dixon, S., Gamache, M., Gomez, M., Grottoli, A.G., Hackerott, S., Hall, C., Karp, R., Olivares-Cordero, D., **Sosa, B.**, The Workshop Consortium, and Kenkel, C.D. Co-designing a future for Caribbean reefs: recommendations for the implementation of advanced coral restoration strategies. *In prep.*

Technical Reports and Policy Briefs

Goldstein, B., & Caribbean Restoration Interventions Forum. (2025). Advancing Coral Reef Interventions in the Caribbean: Overcoming Barriers and Enabling Action. Coral Restoration Consortium. [Policy Brief](#) (**Sosa, B.** listed in Forum)

Duran, A., Avedo, A., Barreira, W., **Sosa, B.**, and Hiaasen, C. (2025). Scaling up mariculture

of a native herbivore in support of "Mission: Iconic Reefs" coral reef restoration. *FDEP Internal Report*, 15 pp.

Cousins, K., Fletcher, A., Hindsley, P., Chabba, M., Lin, A., Needell, A., Ketu, Y., Delaney, G., Aguirre N., **Sosa, B.** (2025). Thriving Everglades, Thriving Economy: Nature's Value in the Everglades. Earth Economics and The Everglades Foundation. *Technical Report*.

PROFESSIONAL PRESENTATIONS

2026 **Sosa, B.**, Jarrett, M., and Butler, M. Quarry-based Mariculture of the Caribbean King Crab in Support of Reef Restoration. Marine Biology Course, Florida Southern College. (Guest Lecture)

2026 **Sosa, B.** Coupled Social-Ecological Research: Florida's Coral Reef Restoration. EVR 6377 - Natural Resource Conservation & Policy, Florida International University. (Guest Lecture)

2026 **Sosa, B.**, and Bhat, M. Restoring Reefs: Public Perceptions and Willingness-to-Pay for Reef Restoration in Florida. McKnight Annual Fellows Meeting, Tampa, FL. (Talk)

2025 **Sosa, B.**, Bhat, M., and Harborne, A. Using Marine Grazers to Facilitate Coral Reef Restoration in the Florida Keys. FKNMS Science Talks, Florida Keys National Marine Sanctuary. (Talk)

2025 **Sosa, B.**, Jarrett, M., Butler, M. Crab Supply for Coral Recovery: Evaluating Quarry-Based Mariculture to Scale Herbivore Restoration. ReefFlorida Symposium 2025, Phillip and Patricia Frost Museum of Science, Miami, FL. (Talk)

2025 **Sosa, B.**, Jarrett, M., and Butler, M. Quarry-based Mariculture of the Caribbean King Crab in Support of Reef Restoration. Florida Keys Marine Science Conference. (Poster)

2025 **Sosa, B.** Discrete choice experiments in reef restoration to understand perceptions and values. EVR 7322 - Methods of Sustainable Resource Management, Florida International University. (Guest Lecture)

2024 **Sosa, B.**, Bhat, M., and Butler, M. Valuing marine grazers as a tool for facilitating coral reef restoration in the Florida Keys. Reef Futures 2024 Conference, Quintana Roo, Mexico. (Talk)

2023-2025 **Sosa, B.** Igniting your research: Crafting questions, bridging gaps. McKnight Fellows' Orientation Meeting, Tampa, Florida. (Invited Talk)

2023 **Sosa, B.**, Bhat, M., and Butler, M. Blueprints for the Blue: Practical Considerations and Economic Strategies of Grazer Stocking for Reef Management in the Florida Keys. International Society of Ecological Economics (ISEE), Santa Marta, Colombia. (Talk)

2023 Vorseth, C., **Sosa, B.**, DeVito, L., and Bhat, M. Valuing Ecosystem Services of Everglades Restoration: Regional and National Policy Implications. Florida Coastal Everglades LTER All Scientists Meeting, Miami, Florida. (Poster)

2023 **Sosa, B.** Blueprints for the Blue: Practical and Economic Considerations for Grazer Stocking Strategies in the Florida Keys. Doctoral Proposal Defense, Florida International University.

GRANTS AND FELLOWSHIPS

Awarded

2024 FDEP Coral Restoration Program, \$198,446 (contributed to proposal; PI: M. Butler)

2023–2025 FEF McKnight Doctoral Fellowship

2021–2023 Presidential Fellowship

2023–2025 Graduate Research Awards, \$20,000+ total (NSF CREST; International Women's Fishing Association; Association of Marine Laboratories of the Caribbean; Coastlines and Oceans)

2019 Benjamin A. Gilman International Scholarship, \$5,000

2017–2020 Various merit-based undergraduate scholarships, over \$70,000

Proposals Under Review or Not Funded

2026 EPA South Florida Program, \$476,029 – *Under Review* (co-author with A. Harborne)

2026 NFWF NCRF Grant, \$928,704 – *Under Review* (contributed to proposal; PI: Kylie Smith)

2024 NOAA Aquaculture Initiative, \$919,767 (contributed to proposal; PI: Butler) – *Not funded*

2023 NMFS-Sea Grant Joint Fellowship Program – *Not funded*

AWARDS AND HONORS

2025 Earth and Environment Symposium - Best Oral Presentation

2024 South Florida Sustainable Environments Internship - Mentor Award

2024 Newfound Harbor Marine Institute, Seacamp - Science Speaker

2023 Florida Coastal Everglades LTER Presentation 2nd Place

2018, 2020 Outstanding Scholar in Forestry & Natural Resources

2017–2020 Dean's List and Semester Honors

2017–2019 National Hispanic Scholar

MENTORSHIP AND TEACHING

2026–Present Hispanic Scholarship Fund (HSF) Mentor

- Former HSF Scholar; now providing 1-on-1 mentorship to current Hispanic undergraduates on academic planning and research career pathways

2023–Present McKnight Doctoral Fellows Mentorship

Florida Education Fund (FEF)

- Recurring invited speaker at the McKnight Doctoral Fellows' Orientation, to incoming fellows on dissertation framing, research question development, and interdisciplinary design
- Provide informal peer mentorship to incoming Hispanic and Black doctoral fellows navigating early-stage graduate school, advisor relationships, and proposal writing

2025 Teaching Assistant, Environmental Science

Florida International University, Department of Earth and Environment

- Led multiple sections (lectures and labs) covering ocean acidification, coastal protection, carbon footprints, and related topics
- Organized and guided field trips and tours of marine restoration labs, connecting

classroom concepts to applied research

2022–Present Undergraduate Research Mentorship

South Florida Sustainable Environments Internship, Florida International University

- Mentored undergraduate interns in data collection, statistical analysis, and scientific presentation and reporting skills

Coastal Conservation and Restoration Lab, Florida International University

- Supervised and trained undergraduate volunteers in animal husbandry and aquacultural construction techniques
- Organized laboratory interns in daily maintenance and experimental protocols

Field Research Training

- Led undergraduate and new graduate students in field methods including SCUBA diving, benthic surveys, coral restoration techniques, and coral outplanting

Mentee Accomplishments:

- Over 15 students mentored, with several presenting research findings at FIU symposia and regional conferences
- Supported multiple undergraduate students in transitioning to graduate programs in marine science
- Co-mentoring doctoral student on input-output economic impact assessment of a marine reserve designation in Biscayne National Park, including fisher and public survey design (mentee: D. Gautam; with M. Bhat, 2025–present)

COMMUNITY OUTREACH AND SCIENCE COMMUNICATION

2022–Present Educational Outreach

ReefBites, ICRS Early Career Chapter (2026–Present)

- Contributing blog writer communicating coral reef science to general audiences

Science Communication and Film

- CISME Film Lead  — NSF IRES 2025 Film Lead  — Miami's Living Seawalls Film Lead

Miami-Dade Elementary and Middle Schools

- Conducted marine science presentations and hands-on activities for K-12 students across Miami-Dade County schools to promote ocean conservation awareness
- Developed age-appropriate educational materials on coral reef ecosystems and restoration efforts
- Career day presenter at Lake Stevens Middle School (2022-2023), introducing students to marine science research pathways

Newfound Harbor Marine Institute - Seacamp, Florida Keys

- Led interactive marine biology sessions for visiting student groups
- Facilitated hands-on reef restoration in-class activities

SKILLS AND CERTIFICATIONS

Field: AAUS Scientific Diver (100+ research dives); FIU Motorboat Crew Chief — vessels to 30 ft (day and night); Coral Restoration & Outplanting; Marine Benthic Surveys; AGRRA Fish Surveys;

Benthic Flux Chambers; Photogrammetry

Software: R, Python, Quarto, LaTeX, R Shiny, GitHub, ArcGIS Pro, NetLogo, Apollo, CoralNet, CPCE, Metashape, Qualtrics, YSI KOR

Analysis & Modeling: Agent-based Modeling, Bioeconomic Modeling, Discrete Choice Experiments (Willingness-to-Pay, Stated Preference Methods), Ecosystem Service Valuation, Cost-Benefit Analysis, Input-Output Modeling, Meta-analysis, Restoration Modeling, Monte Carlo & Sensitivity Analysis, Survey Design, Data Visualization

Laboratory: Aquaculture System Design & Maintenance (flow-through and recirculating); Invertebrate Husbandry (larval rearing, juvenile grow-out, spawning); Stomach Content Analysis; Microplastic Identification; Carbon Estimation; Microscopy; PCR Genomic Analysis; Fish Necropsy

Policy & Communication: Stakeholder engagement, multi-agency coordination, technical report and policy brief writing, plain language science communication, grantsmanship, economic valuation

Professional Development: Fundamentals of Climate Leadership (FIU Institute of Environment × Sachamama, 2025); Environmental Dynamics & Causality Workshop (2025); R Shiny Web Tool Workshop (2025); Python for Environmental Science (2024)

Languages: English (native), Spanish (beginner-spoken)